

Lab 9 — Introduction to Deep Neural Networks using Keras

Aim

To build and train a simple Deep Neural Network (DNN) with multiple hidden layers using TensorFlow/Keras on the MNIST handwritten digit dataset.

Algorithm / Procedure

1. Load the MNIST handwritten digit dataset.
2. Normalize the pixel values from the range 0–255 to 0–1.
3. Create a Sequential neural network.
4. Flatten the 28×28 images into a one-dimensional vector.
5. Add two hidden layers with 128 and 64 neurons using ReLU activation.
6. Add an output layer with 10 neurons using Softmax activation.
7. Compile the model using the Adam optimizer and sparse categorical cross-entropy loss.
8. Train the model for 5 epochs.
9. Evaluate the trained model on the test dataset.
10. Display the test accuracy.

Program

```
import tensorflow as tf
from tensorflow.keras import layers, models

# Load MNIST dataset
(X_train, y_train), (X_test, y_test) = tf.keras.datasets.mnist.load_data()

# Normalize pixel values
X_train = X_train / 255.0
X_test = X_test / 255.0

# Create Deep Neural Network
model = models.Sequential([
    layers.Input(shape=(28, 28)),
    layers.Flatten(),
    layers.Dense(128, activation='relu'),
    layers.Dense(64, activation='relu'),
    layers.Dense(10, activation='softmax')
])

# Compile model
model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

# Train model
model.fit(
    X_train,
    y_train,
    epochs=5,
    validation_split=0.1
)

# Evaluate model
test_loss, test_accuracy = model.evaluate(X_test, y_test)

print("Test accuracy:", test_accuracy)
```

OUTPUT

```
... Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11490434/11490434 ————— 0s 0us/step
Epoch 1/5
1688/1688 ————— 13s 6ms/step - accuracy: 0.9246 - loss: 0.2559 - val_accuracy: 0.9697 - val_loss: 0.1042
Epoch 2/5
1688/1688 ————— 8s 5ms/step - accuracy: 0.9673 - loss: 0.1045 - val_accuracy: 0.9752 - val_loss: 0.0898
Epoch 3/5
1688/1688 ————— 7s 4ms/step - accuracy: 0.9776 - loss: 0.0714 - val_accuracy: 0.9715 - val_loss: 0.0984
Epoch 4/5
1688/1688 ————— 8s 5ms/step - accuracy: 0.9816 - loss: 0.0566 - val_accuracy: 0.9773 - val_loss: 0.0742
Epoch 5/5
1688/1688 ————— 8s 5ms/step - accuracy: 0.9861 - loss: 0.0428 - val_accuracy: 0.9757 - val_loss: 0.0918
313/313 ————— 1s 4ms/step - accuracy: 0.9734 - loss: 0.0818
Test accuracy: 0.9733999967575073
```

RESULT

A Deep Neural Network with two hidden layers was successfully built and trained using TensorFlow/Keras on the MNIST dataset. The model achieved approximately 97% test accuracy in recognizing handwritten digits.